

# Michael Rizzo Smith

## Curriculum Vitae

Nashville, TN — michael.smith.4@vanderbilt.edu — Website

### EDUCATION

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| <b>Ph.D Astrophysics</b><br>Vanderbilt University<br>Department of Physics and Astronomy<br><i>Advisor: Kelly Holley-Bockelmann</i>                                                      | Aug. 2023 - Exp. 2029<br>Nashville, TN  |
| <b>Bachelor of Science</b> , Physics and Astronomy & Planetary Science<br>Stony Brook University<br>Department of Physics and Astronomy<br><i>Magna Cum Laude with Honors in Physics</i> | Aug. 2017 - May 2021<br>Stony Brook, NY |

### AWARDS AND FELLOWSHIPS

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| <b>NSF Graduate Research Fellow</b> (\$150,000)  | Jun. 2025            |
| Vanderbilt Graduate School Travel Grant          | July 2024            |
| McMinn Research Award                            | May 2024             |
| Vanderbilt Robert T. Lagemann Award              | Apr. 2024            |
| Departmental Honors, Dept. of Physics            | May 2021             |
| <b>Stony Brook Campus Scholarship</b> (\$26,000) | Aug. 2017 - May 2021 |

### PROFESSIONAL WORK AND AFFILIATIONS

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| <b>NASA XRP 2026 Review Panel</b> , Executive Secretary                                                                            | Mar. 2026             |
| <b>LISA Consortium</b> , Core Member                                                                                               | May 2025 - Present    |
| <b>Vanderbilt's Graduate Student Astrophysics Journal Research Club</b> ,<br>Co-Coordinator                                        | Jan. 2025 - Present   |
| Rebuilt existing club structure to emphasize practical presentation skills and implemented<br>a formal faculty feedback structure. |                       |
| <b>LISA Disc-IMRI Project Astrophysics Working Group</b> , Code Captain                                                            | Dec. 2023 - Dec. 2025 |
| <b>Establishing Multimessenger Astronomy Inclusive Training (EMIT)</b> ,<br>Graduate Student Scholar                               | Sep. 2023 - Present   |
| <b>ASAS-SN Project</b> , Data Analyst                                                                                              | Jun. 2021 - July 2023 |
| Research Assistant for the All Sky Automated Survey for Supernovae<br>at The Ohio State University.                                |                       |

### OUTREACH

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| <b>Astronomy on Tap Nashville</b> , Host                                                                                                                       | 2025 - Present |
| <b>10-Years of Gravitational Waves, Vanderbilt Community Celebration</b> , Poster<br><i>"It's OK to Be Gassy: How Cosmic Gas Guides Black Hole Collisions"</i> | Sep. 2025      |
| <b>NASA Advocacy Day</b> , Vanderbilt Federal Relations<br>Met with congressional staff in Washington D.C. to emphasize funding public science and NASA.       | July 2025      |
| <b>NSF GRFP: Navigating the Application Panel</b> , Panelist<br>Panel to prospective applicants.                                                               | Oct. 2025      |
| <b>Vanderbilt Society of Physics Students</b> , Invited Panelist<br>Panel on graduate school application process.                                              | Oct. 2025      |
| <b>EMIT Outreach</b> , Invited Speaker<br>Career talk on computational astrophysics and higher ed.                                                             | Jan. 2025      |
| <b>Vanderbilt Society of Physics Students</b> , Invited Panelist<br>Panel on graduate school application process.                                              | Oct. 2024      |
| <b>Valor Collegiate Academy</b> , Invited Speaker<br>Career talk to first-year class.                                                                          | Mar. 2024      |

## TEACHING

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### Graduate Teaching Assistant

Fall 2023 - Spring 2025

Course Title: ASTR 1010L: Intro to Astronomy Lab

Number of Students: ~21 per term

### Private Tutor

Feb. 2020 - Oct. 2020

Intro Physics for Medical Professionals

### Undergraduate Teaching Assistant

Spring 2020

Course Title: AST 203: Astronomy

## PRESENTATIONS

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### Contributed Talk, Vanderbilt's Astrophysics Journal Research Club

Nov. 2025

Presenting: *"Relativistic Binary Precession: Impact on Eccentric Massive Binary Black Hole Accretion and Hydrodynamics"*

### Contributed Talk, Vanderbilt's Astrophysics Journal Research Club

Jan. 2025

Presenting: *"The dynamics and electromagnetic signatures of accretion in unequal mass binary black hole inspirals"*

### Contributed Talk, The 15th International LISA Symposium

July 2024

Presenting on behalf of the Disc-IMRI Code Comparison LISA Astrophysics Working Group

## PUBLICATIONS

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### Reviewed Articles

(1 first-author, 5 co-author, 141 citations)

1. Andrea Derdzinski, Alexander J. Dittmann, Alessia Franchini, Alessandro Lupi, Noé Brucy, Pedro R. Capelo, Frédéric S. Masset, Raphaël Mignon-Risse, **Michael Rizzo Smith**, Edwin Santiago-Leandro, Martina Toscani, David A. Velasco-Romero, Robert Wissing, Mudit Garg, Lucio Mayer, Roberto Serafinelli, Lazaros Souvatzis, Daniel J. D'Orazio, and Jonathan Menu. "The LISA Astrophysics "Disc-IMRI" Code Comparison Project: Intermediate-Mass-Ratio Binaries in AGN-Like Discs". In: *arXiv e-prints*, arXiv:2512.10893 (Dec. 2025), arXiv:2512.10893. arXiv: 2512.10893 [astro-ph.HE]
2. Kelly Holley-Bockelmann, Fazeel Khan, Isaiah Williams, Jaelyn Roth, **Michael Rizzo Smith**, Kaitlin Porter, Jillian Bellovary, Andrea Derdzinski, and Andrea Macciò. "Handy Relation Between Binary Black Hole Merger Times and Host Galaxy Properties". In: *arXiv e-prints*, arXiv:2508.14253 (Aug. 2025), arXiv:2508.14253. arXiv: 2508.14253 [astro-ph.GA]
3. J. M. M. Neustadt, C. S. Kochanek, and **Rizzo Smith, M.** "Constraints on pre-SN outbursts from the progenitor of SN 2023ixf using the large binocular telescope". In: *MNRAS* 527.3 (Jan. 2024), pp. 5366–5373. arXiv: 2306.06162 [astro-ph.HE]
4. Matthew Kenworthy, [20 authors], and **Rizzo Smith, Michael.** "A planetary collision afterglow and transit of the resultant debris cloud". In: *Nature* 622.7982 (Oct. 2023), pp. 251–254. arXiv: 2310.08360 [astro-ph.EP]
5. **Rizzo Smith, M.**, C. S. Kochanek, and J. M. M. Neustadt. "The late time optical evolution of twelve core-collapse supernovae: detection of normal stellar winds". In: *MNRAS* 523.1 (July 2023), pp. 1474–1495. arXiv: 2212.09763 [astro-ph.HE]
6. A. Kawash, [11 authors], **Rizzo Smith, M.**, T. W. -S. Holoién, J. L. Prieto, and T. A. Thompson. "The Galactic Nova Rate: Estimates from the ASAS-SN and Gaia Surveys". In: *The Astrophysical Journal* 937.2, 64 (Oct. 2022), p. 64. arXiv: 2206.14132

### Non-Reviewed Articles

(6 first-author, 13 citations)

1. M. Rizzo Smith and ASASSN-Team. "An Update on ASASSN-21qj: A Rapidly Fading, Sun-Like Star; Back With a Vengeance". In: *The Astronomer's Telegram* 15531 (July 2022), p. 1
2. M. Rizzo Smith and ASASSN-Team. "ASASSN-22el: A Deep Eclipse Event". In: *The Astronomer's Telegram* 15308 (Apr. 2022), p. 1
3. M. Rizzo Smith and ASASSN-Team. "ASASSN-21sa: A Deep Dimming Event". In: *The Astronomer's Telegram* 14937 (Sept. 2021), p. 1
4. M. Rizzo Smith and ASASSN-Team. "ASASSN-21qj: A Rapidly Fading, Sun-Like Star". In: *The Astronomer's Telegram* 14879 (Aug. 2021), p. 1
5. M. Rizzo Smith and ASASSN-Team. "ASASSN-21nn: An Unusual Dimming Event in a Red Giant Star". In: *The Astronomer's Telegram* 14803 (July 2021), p. 1

6. M. Rizzo Smith and ASASSN-Team. “ASASSN-21ml: Discovery of an Extreme ( $\Delta g > 10$  mag) L-Dwarf Flare”. In: *The Astronomer’s Telegram* 14778 (July 2021), p. 1